Pikachu Metrics 数据分析 metrics_walk_forward_20260820_194147_with_target

FILE
metrics_walk_forward_20260820_194147_with_target.csv

NETWORK
TransitionUp

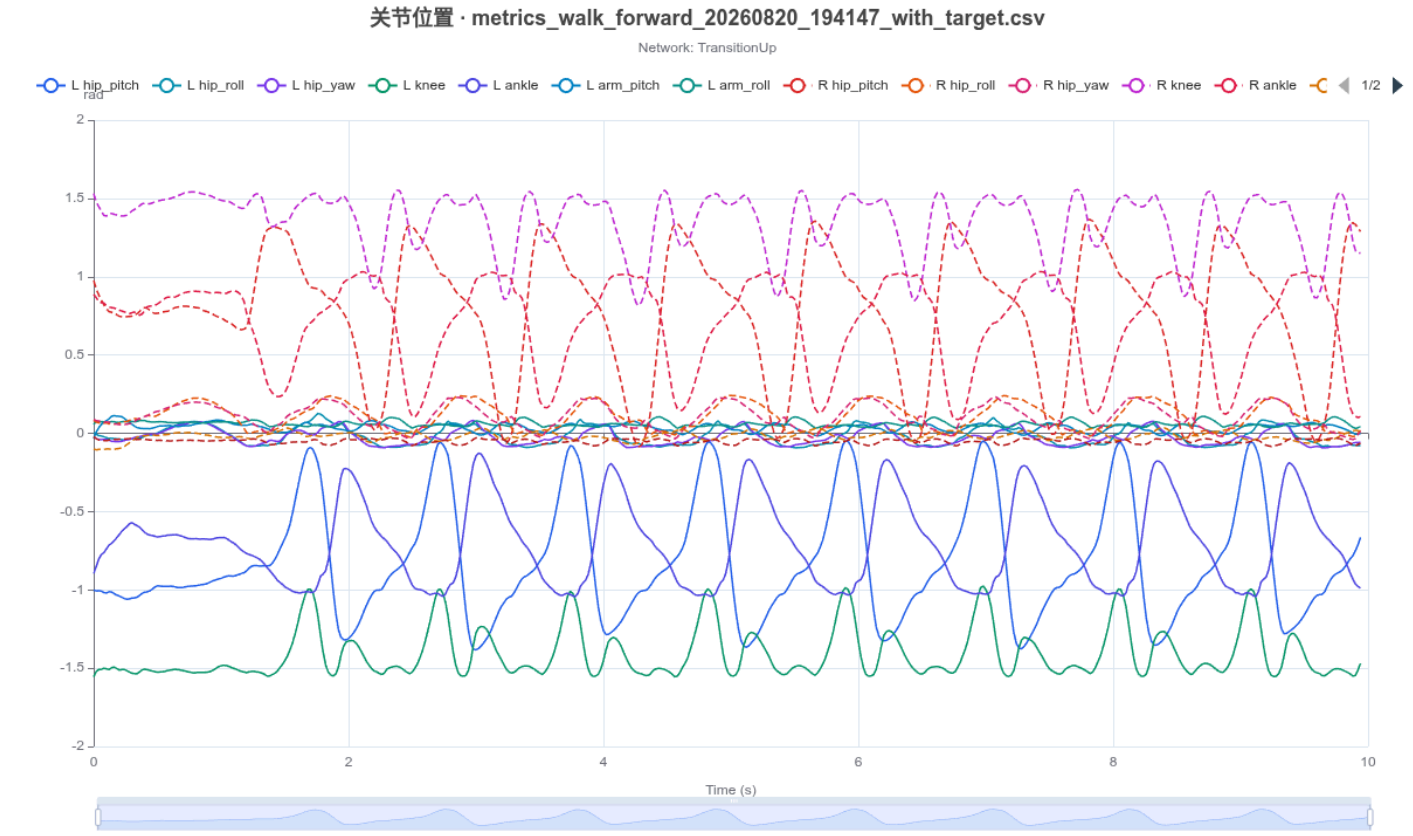
SAMPLES
498

DURATION
9.940 s

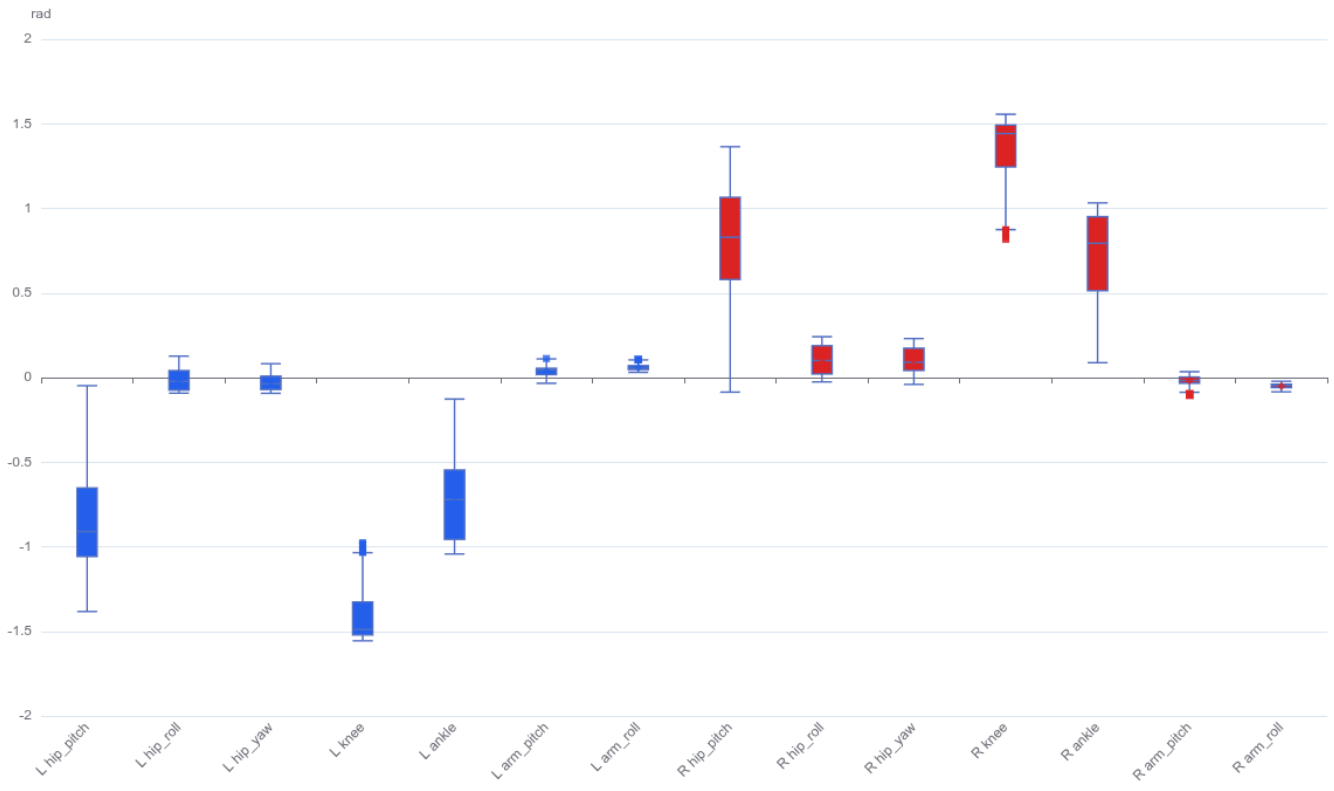
ENVIRONMENTS
1

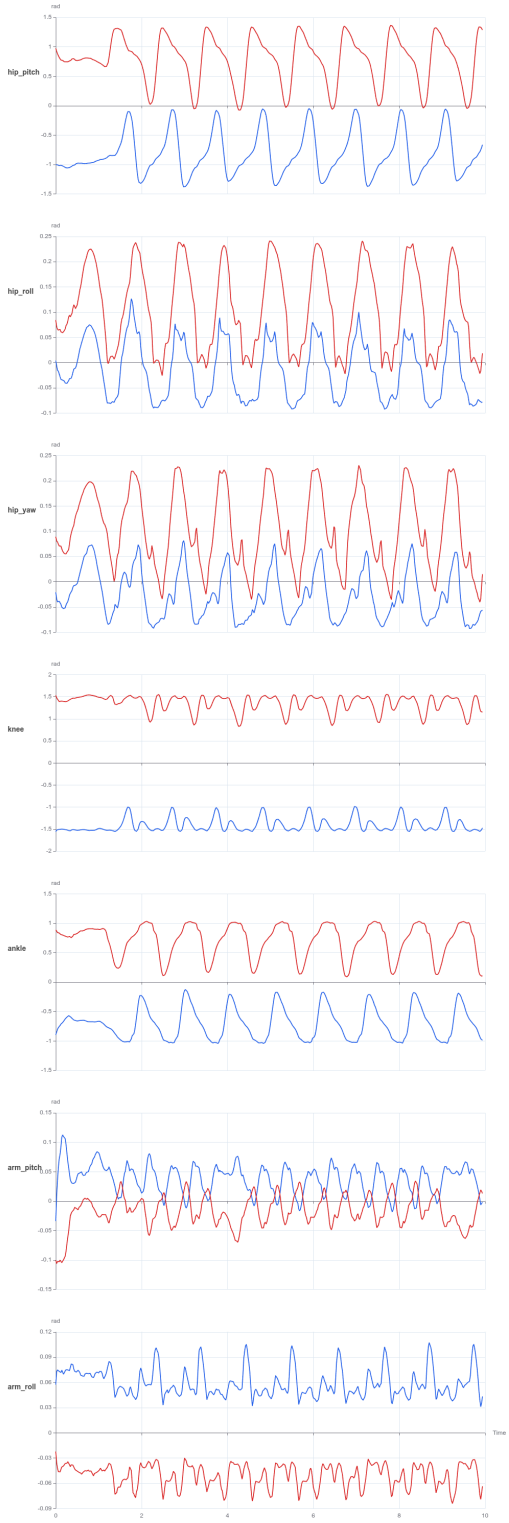
时间区间: 0.000 — 9.940 s · 仅含已勾选字段

关节位置

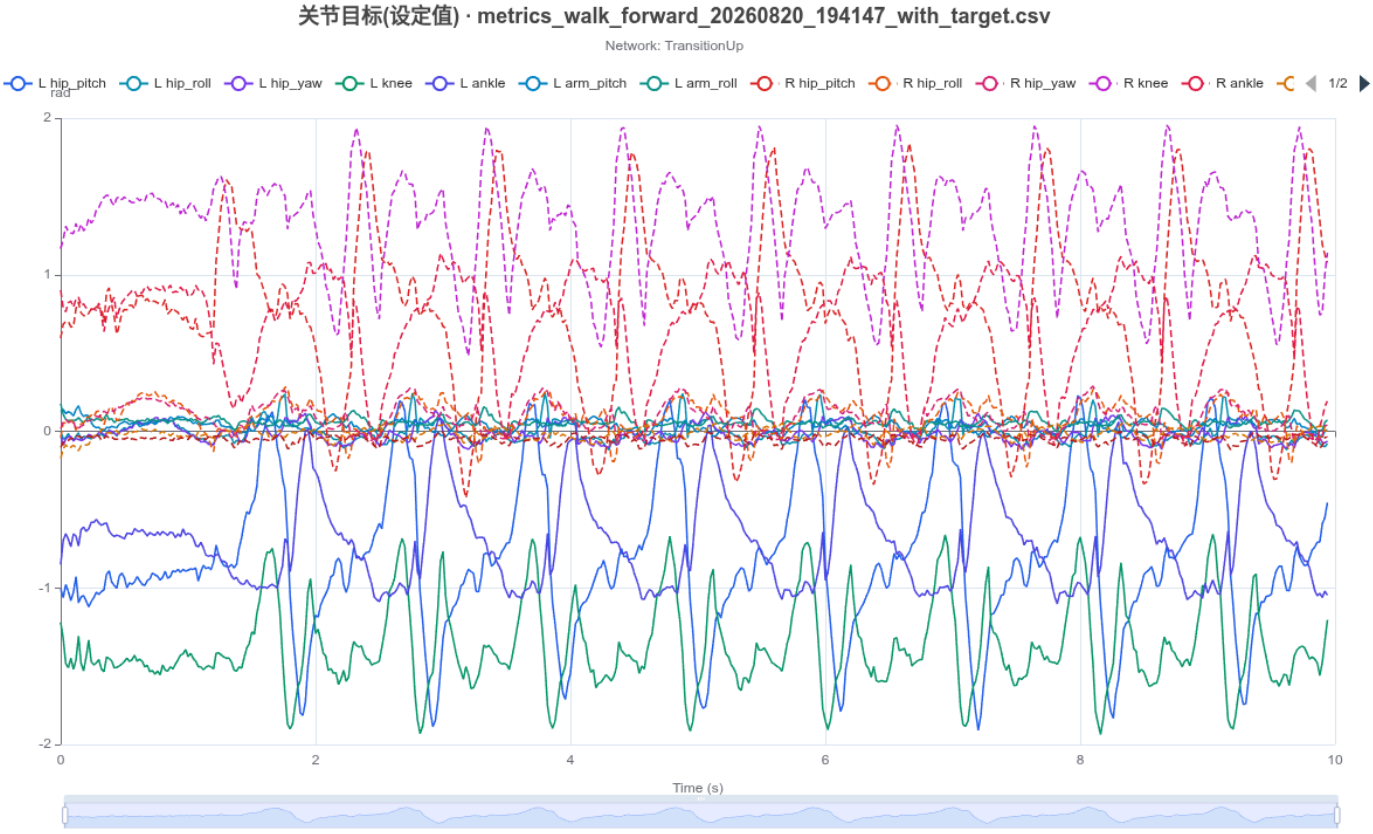


Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
joint_position_rad/left_hip_pitch_joint	30	10	498	-1.38278	-0.0482672	-0.826908	0.904144	1.38278	1.31825
joint_position_rad/left_hip_roll_joint	10	15	498	-0.0922736	0.126643	-0.0162064	0.0598712	0.126643	0.0887879
joint_position_rad/left_hip_yaw_joint	10	15	498	-0.0929997	0.0814879	-0.0272087	0.0554260	0.0929997	0.0861769
joint_position_rad/left_knee_joint	30	10	498	-1.55591	-0.975935	-1.40895	1.41755	1.55591	1.54663
joint_position_rad/left_ankle_joint	10	15	498	-1.04268	-0.127211	-0.704808	0.752828	1.04268	1.02719
joint_position_rad/left_arm_pitch_joint	10	15	498	-0.0338010	0.112535	0.0373047	0.0448246	0.112535	0.0723762
joint_position_rad/left_arm_roll_joint	10	15	498	0.0310933	0.107308	0.0596236	0.0618784	0.107308	0.0967433
joint_position_rad/right_hip_pitch_joint	30	10	498	-0.0863542	1.36440	0.777837	0.875331	1.36440	1.31952
joint_position_rad/right_hip_roll_joint	10	15	498	-0.0256462	0.241261	0.106571	0.135701	0.241261	0.232424
joint_position_rad/right_hip_yaw_joint	10	15	498	-0.0404367	0.230869	0.102880	0.128651	0.230869	0.221771
joint_position_rad/right_knee_joint	30	10	498	0.816205	1.55627	1.35453	1.36765	1.55627	1.53182
joint_position_rad/right_ankle_joint	10	15	498	0.0875370	1.03210	0.708137	0.763619	1.03210	1.01723
joint_position_rad/right_arm_pitch_joint	10	15	498	-0.106097	0.0346720	-0.0169525	0.0315923	0.106097	0.0589531
joint_position_rad/right_arm_roll_joint	10	15	498	-0.0840285	-0.0222484	-0.0504958	0.0521834	0.0840285	0.0756599

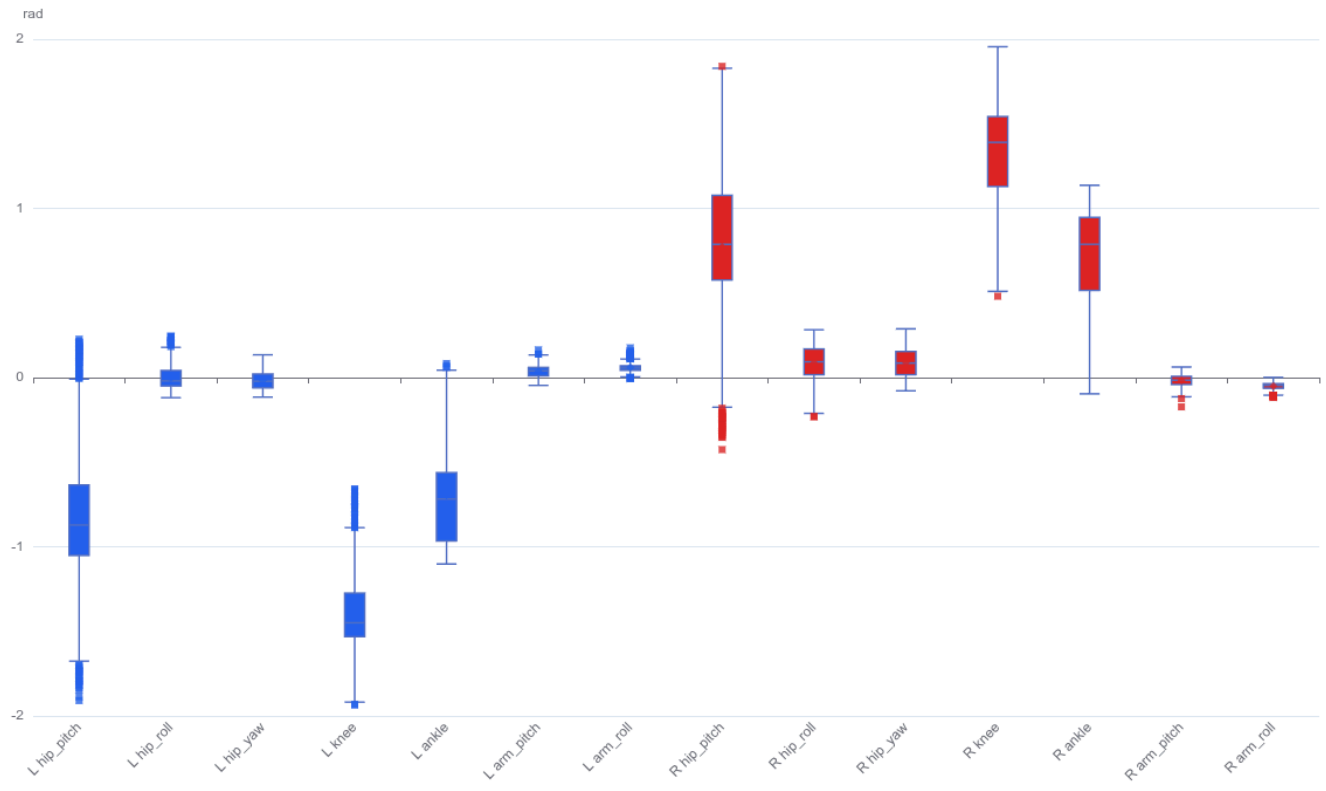

rad
2
1.5
1
0.5
0
-0.5
-1
-1.5
-2
L hip_pitch
L hip_roll
L hip_yaw
L knee
L ankle
L arm_pitch
L arm_roll
R hip_pitch
R hip_roll
R hip_yaw
R knee
R ankle
R arm_pitch
R arm_roll

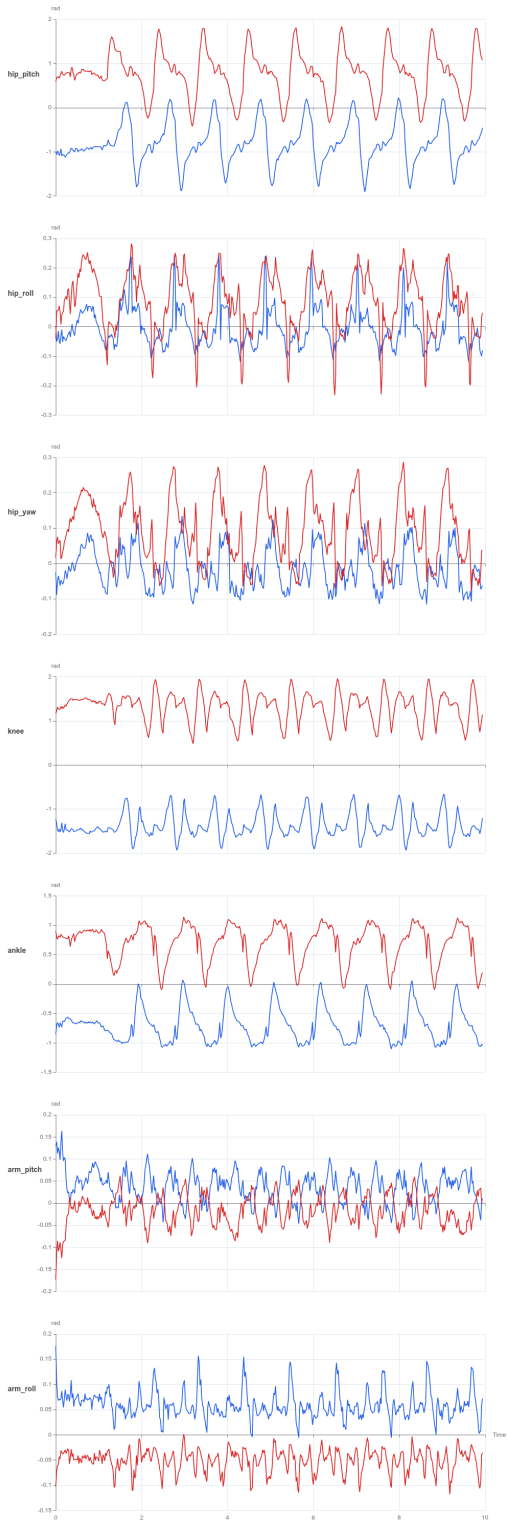

hip_pitch
hip_roll
hip_yaw
knee
ankle
arm_pitch
arm_roll
rad
Time

关节目标(设定值)

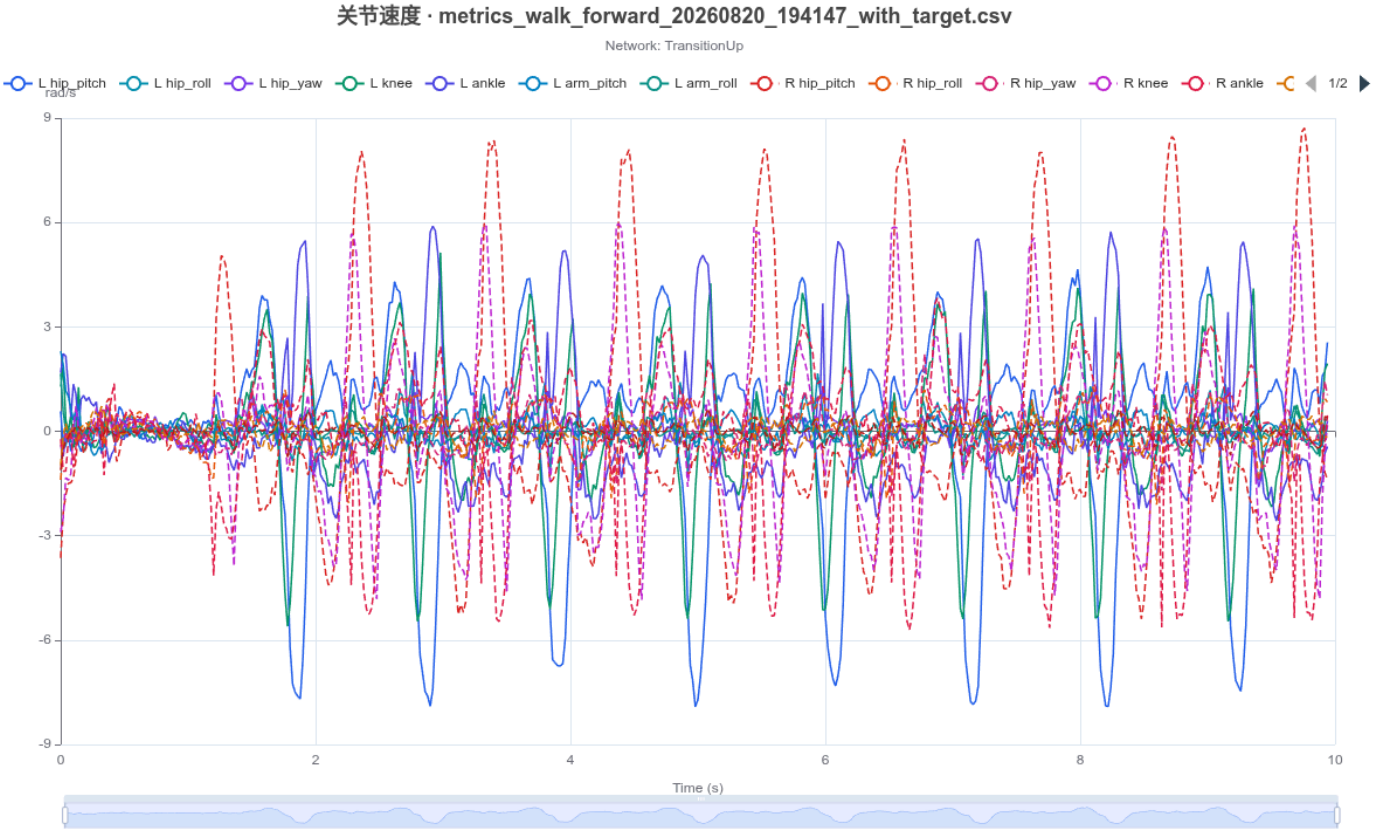


Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
joint_target_rad/left_hip_pitch_joint	30	10	498	-1.90934	0.226845	-0.816674	0.941919	1.90934	1.66536
joint_target_rad/left_hip_roll_joint	10	15	498	-0.119244	0.245389	-0.00341032	0.0685779	0.245389	0.118043
joint_target_rad/left_hip_yaw_joint	10	15	498	-0.115769	0.134540	-0.0158468	0.0586751	0.134540	0.100295
joint_target_rad/left_knee_joint	30	10	498	-1.93612	-0.657720	-1.37622	1.40305	1.93612	1.78596
joint_target_rad/left_ankle_joint	10	15	498	-1.10215	0.0828107	-0.696286	0.756235	1.10215	1.04983
joint_target_rad/left_arm_pitch_joint	10	15	498	-0.0463652	0.163582	0.0368468	0.0508065	0.163582	0.0914360
joint_target_rad/left_arm_roll_joint	10	15	498	-0.00579054	0.175802	0.0598126	0.0657666	0.175802	0.119548
joint_target_rad/right_hip_pitch_joint	30	10	498	-0.425664	1.83939	0.773093	0.932096	1.83939	1.69948
joint_target_rad/right_hip_roll_joint	10	15	498	-0.232236	0.282303	0.0868576	0.133953	0.282303	0.236509
joint_target_rad/right_hip_yaw_joint	10	15	498	-0.0779605	0.287988	0.0892810	0.127225	0.287988	0.251580
joint_target_rad/right_knee_joint	30	10	498	0.480680	1.95569	1.32252	1.36231	1.95569	1.80865
joint_target_rad/right_ankle_joint	10	15	498	-0.0959676	1.13663	0.697585	0.767978	1.13663	1.06662
joint_target_rad/right_arm_pitch_joint	10	15	498	-0.173172	0.0622355	-0.0169733	0.0383945	0.173172	0.0705827
joint_target_rad/right_arm_roll_joint	10	15	498	-0.117793	0.00100856	-0.0510726	0.0558660	0.117793	0.0984786

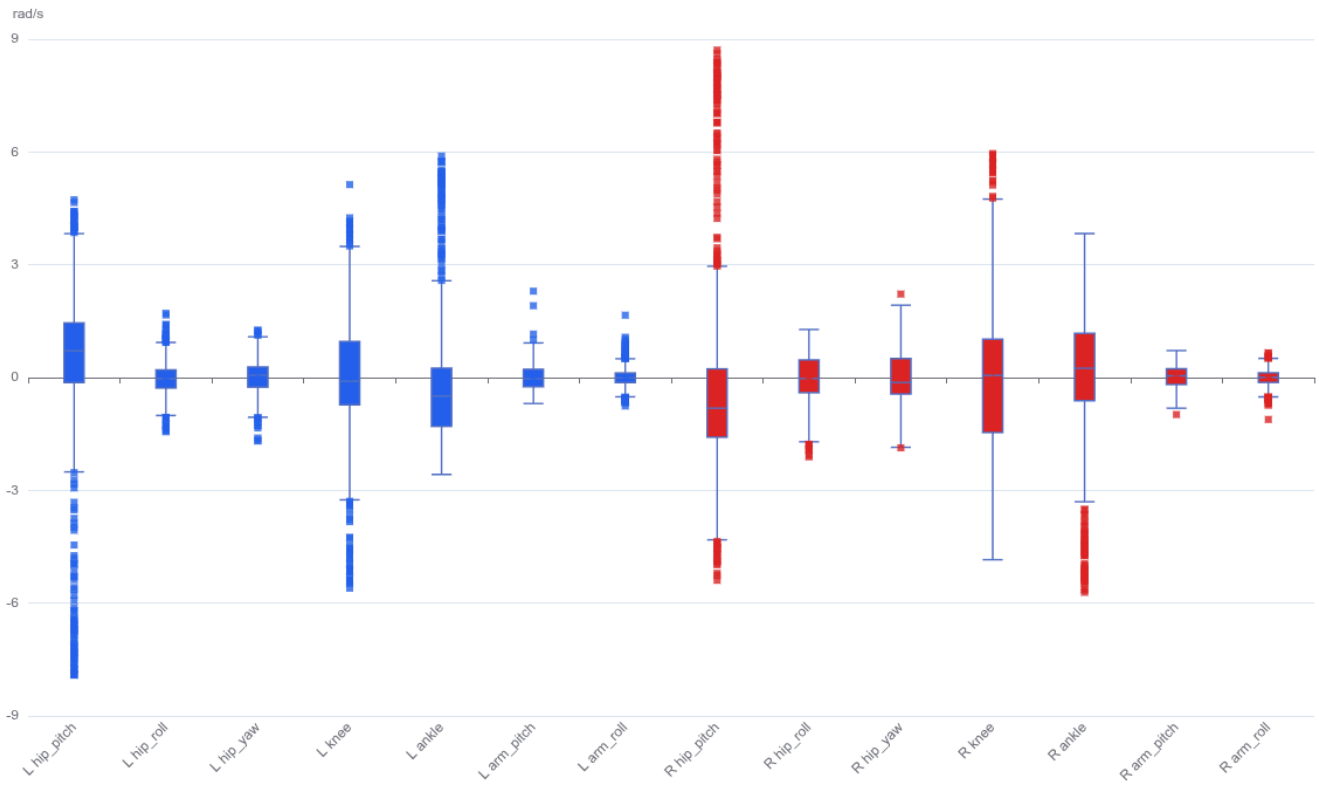

rad
2
1
0
-1
-2
L hip_pitch
L hip_roll
L hip_yaw
L knee
L ankle
L arm_pitch
L arm_roll
R hip_pitch
R hip_roll
R hip_yaw
R knee
R ankle
R arm_pitch
R arm_roll

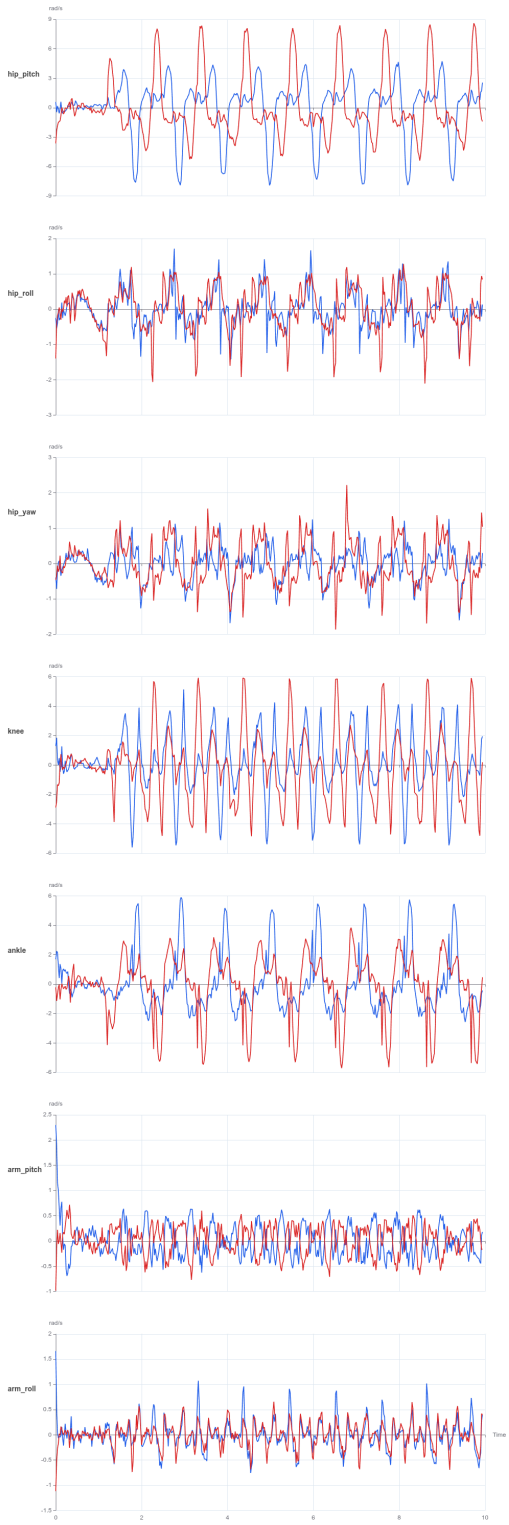

hip_pitch
hip_roll
hip_yaw
knee
ankle
arm_pitch
arm_roll
rad
Time

关节速度

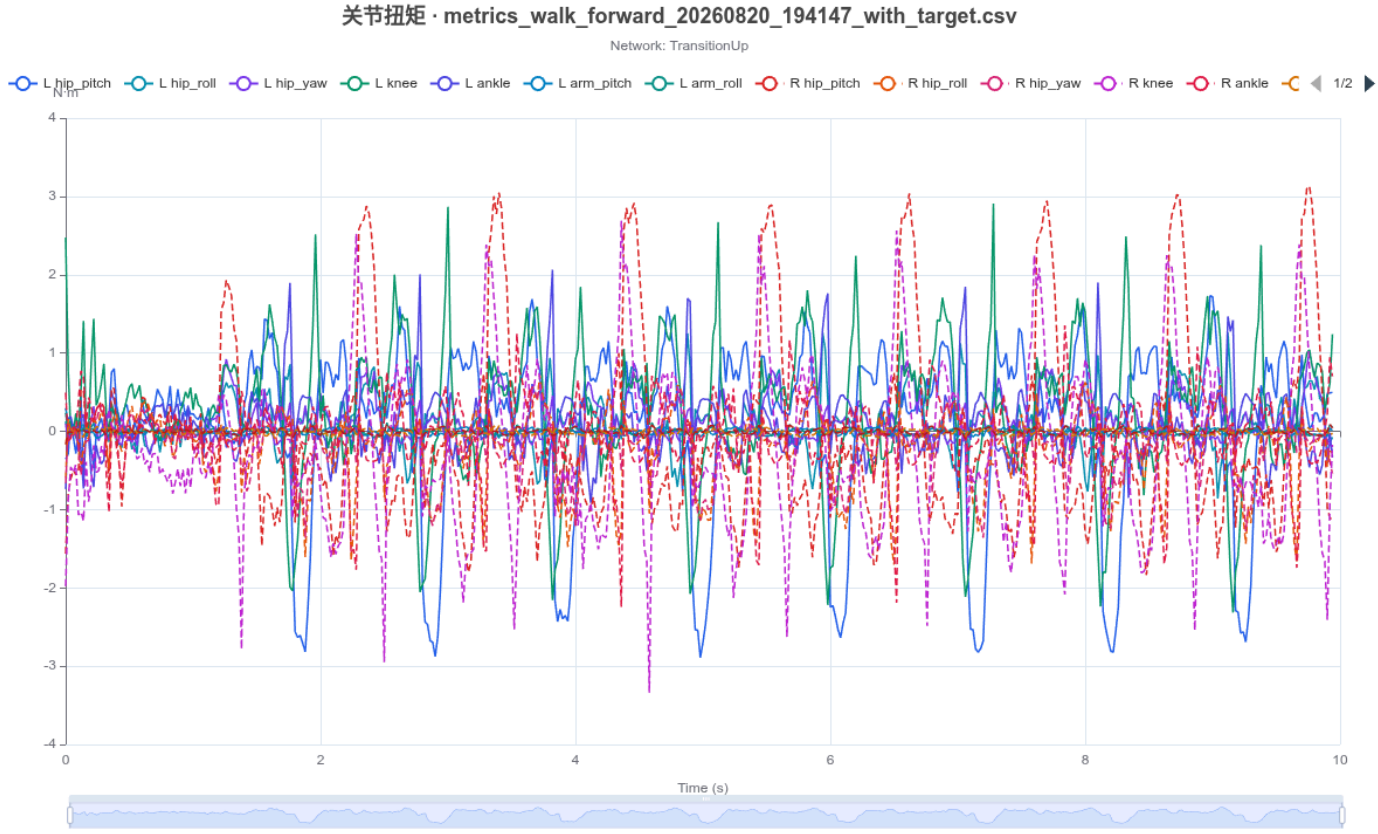


Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
joint_velocity_rad_s/left_hip_pitch_joint	30	10	498	-7.91624	4.72936	0.0347930	2.88514	7.91624	7.06639
joint_velocity_rad_s/left_hip_roll_joint	10	15	498	-1.44097	1.71068	-0.0194482	0.483738	1.71068	1.05700
joint_velocity_rad_s/left_hip_yaw_joint	10	15	498	-1.68712	1.26353	-0.00791968	0.449938	1.68712	0.935871
joint_velocity_rad_s/left_knee_joint	30	10	498	-5.60234	5.13224	0.00744359	1.94501	5.60234	4.24863
joint_velocity_rad_s/left_ankle_joint	10	15	498	-2.57692	5.89308	-0.0154219	1.93939	5.89308	4.88810
joint_velocity_rad_s/left_arm_pitch_joint	10	15	498	-0.691466	2.29832	0.00598656	0.334871	2.29832	0.570766
joint_velocity_rad_s/left_arm_roll_joint	10	15	498	-0.755996	1.65865	0.00171910	0.284807	1.65865	0.599374
joint_velocity_rad_s/right_hip_pitch_joint	30	10	498	-5.39335	8.71084	0.0152717	3.16276	8.71084	7.60261
joint_velocity_rad_s/right_hip_roll_joint	10	15	498	-2.10765	1.27849	-0.00551757	0.639888	2.10765	1.27849
joint_velocity_rad_s/right_hip_yaw_joint	10	15	498	-1.86896	2.22108	-0.00691002	0.623357	2.22108	1.18193
joint_velocity_rad_s/right_knee_joint	30	10	498	-4.84459	5.95915	-0.0369127	2.30775	5.95915	4.81955
joint_velocity_rad_s/right_ankle_joint	10	15	498	-5.71162	3.83051	-0.0808823	2.08415	5.71162	4.95096
joint_velocity_rad_s/right_arm_pitch_joint	10	15	498	-0.983233	0.718144	0.00931497	0.286293	0.983233	0.531345
joint_velocity_rad_s/right_arm_roll_joint	10	15	498	-1.11452	0.657270	-0.00561704	0.243465	1.11452	0.490675

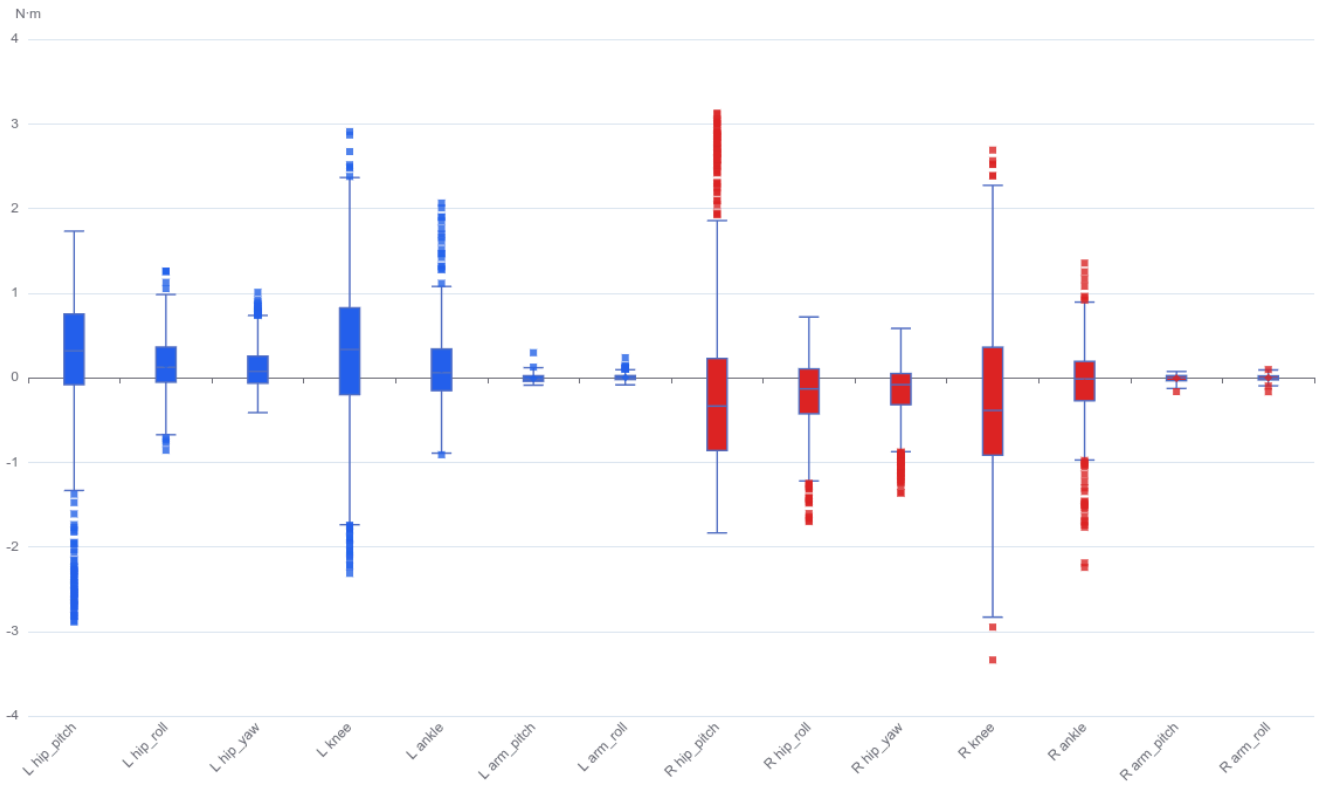

rad/s
9
6
3
0
-3
-6
-9
L hip_pitch
L hip_roll
L hip_yaw
L knee
L ankle
L arm_pitch
L arm_roll
R hip_pitch
R hip_roll
R hip_yaw
R knee
R ankle
R arm_pitch
R arm_roll

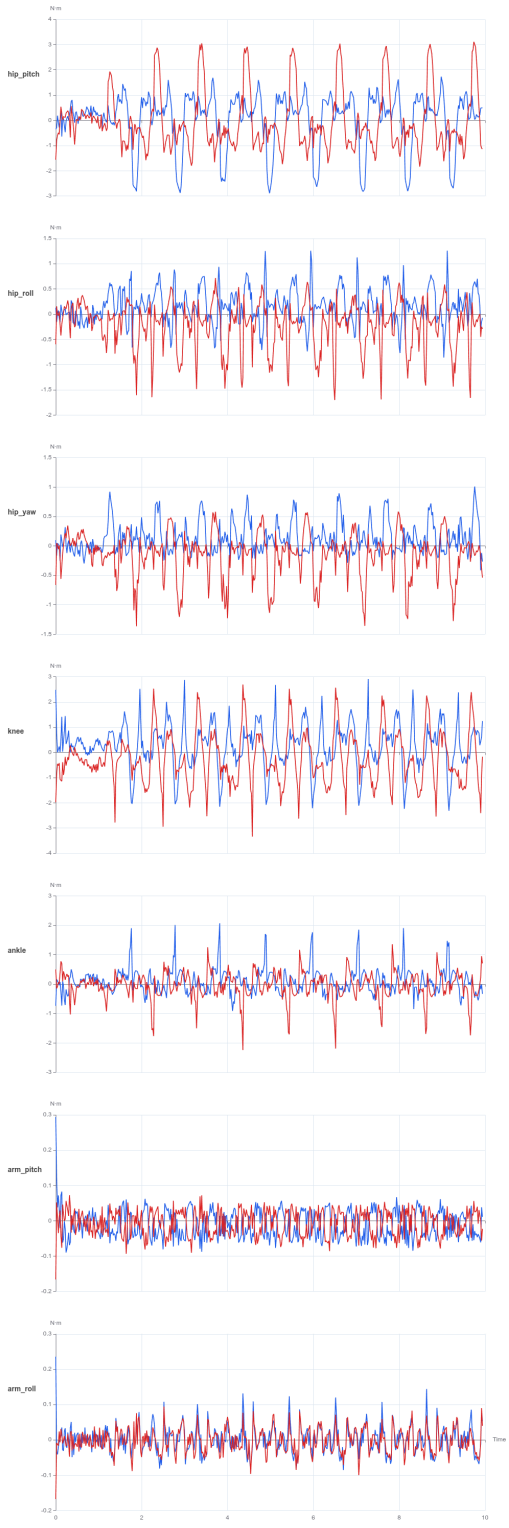

hip_pitch
hip_roll
hip_yaw
knee
ankle
arm_pitch
arm_roll
rad/s
Time

关节扭矩

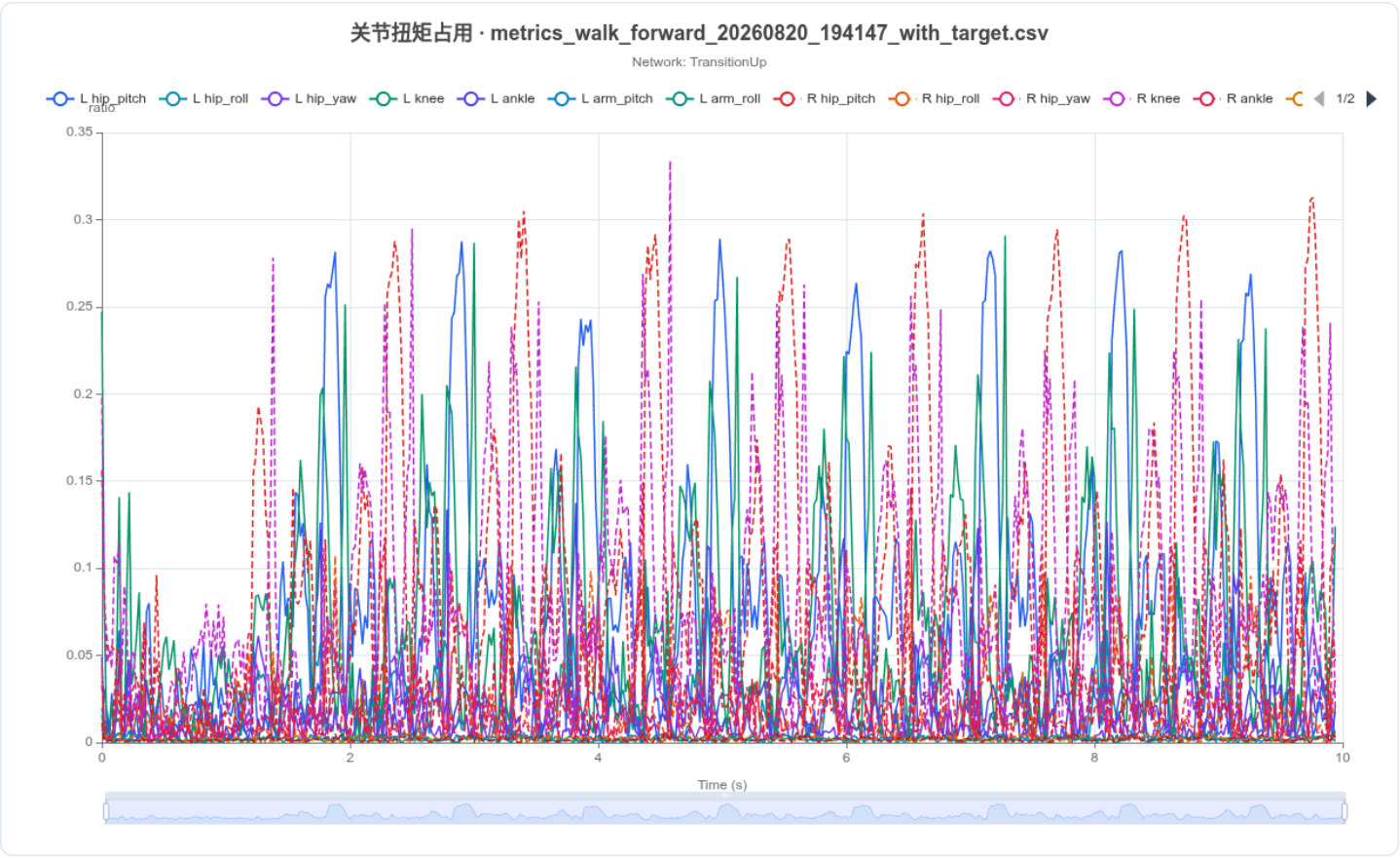


Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
joint_torque_nm/left_hip_pitch_joint	30	10	498	-2.88988	1.73036	0.0700725	1.05998	2.88988	2.53498
joint_torque_nm/left_hip_roll_joint	10	15	498	-0.859629	1.25801	0.154756	0.374444	1.25801	0.755995
joint_torque_nm/left_hip_yaw_joint	10	15	498	-0.415013	1.00809	0.134796	0.295647	1.00809	0.717614
joint_torque_nm/left_knee_joint	30	10	498	-2.31472	2.90768	0.279646	0.920901	2.90768	1.80397
joint_torque_nm/left_ankle_joint	10	15	498	-0.912069	2.06298	0.104343	0.439452	2.06298	0.738994
joint_torque_nm/left_arm_pitch_joint	10	15	498	-0.0898757	0.293927	-0.0101483	0.0413440	0.293927	0.0669061
joint_torque_nm/left_arm_roll_joint	10	15	498	-0.0866058	0.235566	0.000988050	0.0387029	0.235566	0.0700458
joint_torque_nm/right_hip_pitch_joint	30	10	498	-1.83624	3.12709	-0.0527374	1.17194	3.12709	2.71907
joint_torque_nm/right_hip_roll_joint	10	15	498	-1.70247	0.716256	-0.221257	0.519854	1.70247	1.15497
joint_torque_nm/right_hip_yaw_joint	10	15	498	-1.36598	0.581399	-0.153186	0.407134	1.36598	0.934613
joint_torque_nm/right_knee_joint	30	10	498	-3.33822	2.68895	-0.255383	1.04035	3.33822	2.08324
joint_torque_nm/right_ankle_joint	10	15	498	-2.24368	1.35332	-0.0645834	0.469894	2.24368	1.01376
joint_torque_nm/right_arm_pitch_joint	10	15	498	-0.166086	0.0723749	-0.00675483	0.0368543	0.166086	0.0631243
joint_torque_nm/right_arm_roll_joint	10	15	498	-0.167419	0.0946622	-0.00262294	0.0349680	0.167419	0.0669011

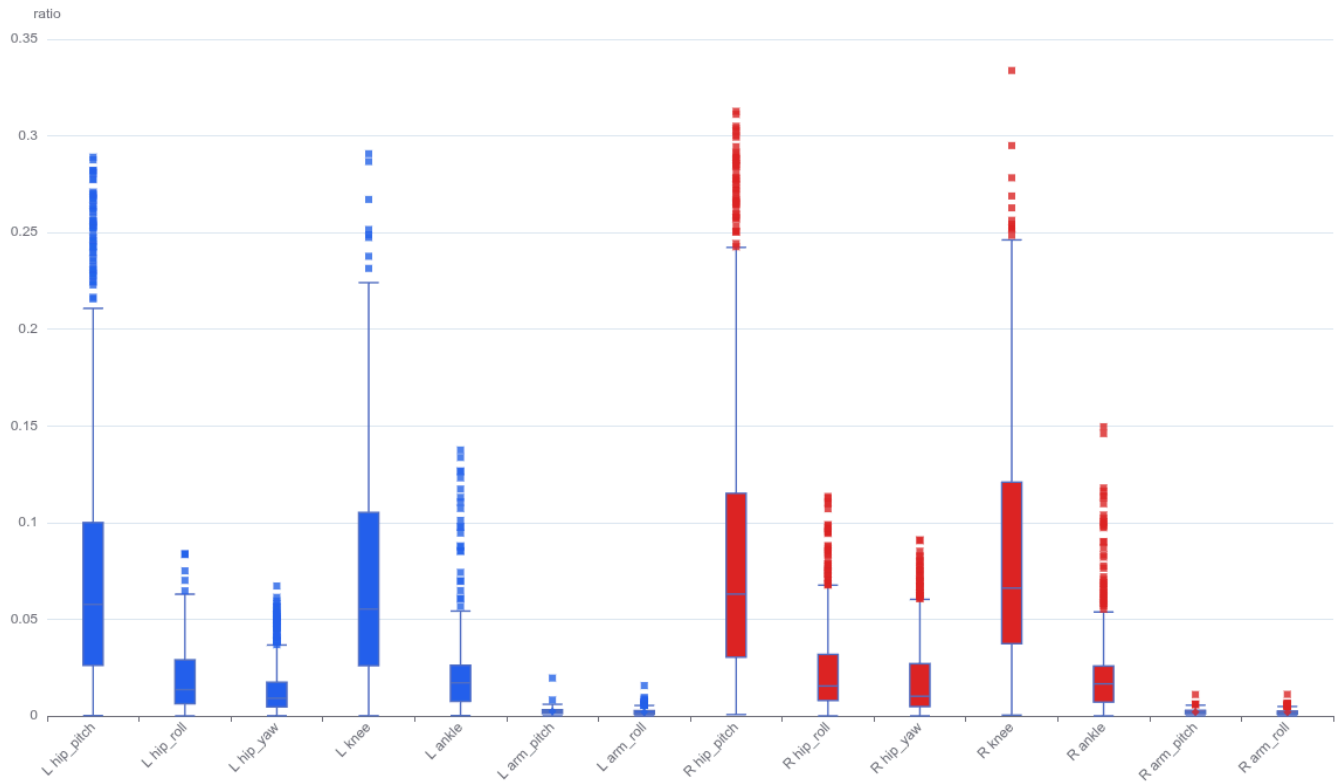

N·m
4
3
2
1
0
-1
-2
-3
-4
L hip_pitch
L hip_roll
L hip_yaw
L knee
L ankle
L arm_pitch
L arm_roll
R hip_pitch
R hip_roll
R hip_yaw
R knee
R ankle
R arm_pitch
R arm_roll

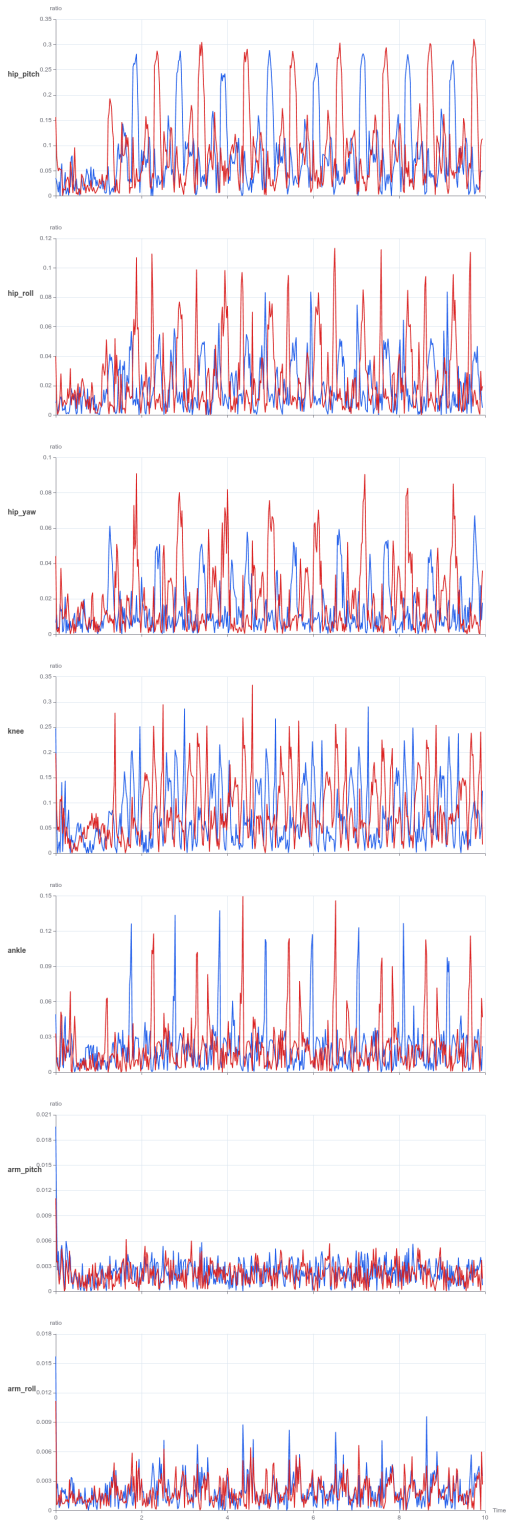

hip_pitch
hip_roll
hip_yaw
knee
ankle
arm_pitch
arm_roll
N·m
Time

关节扭矩占用

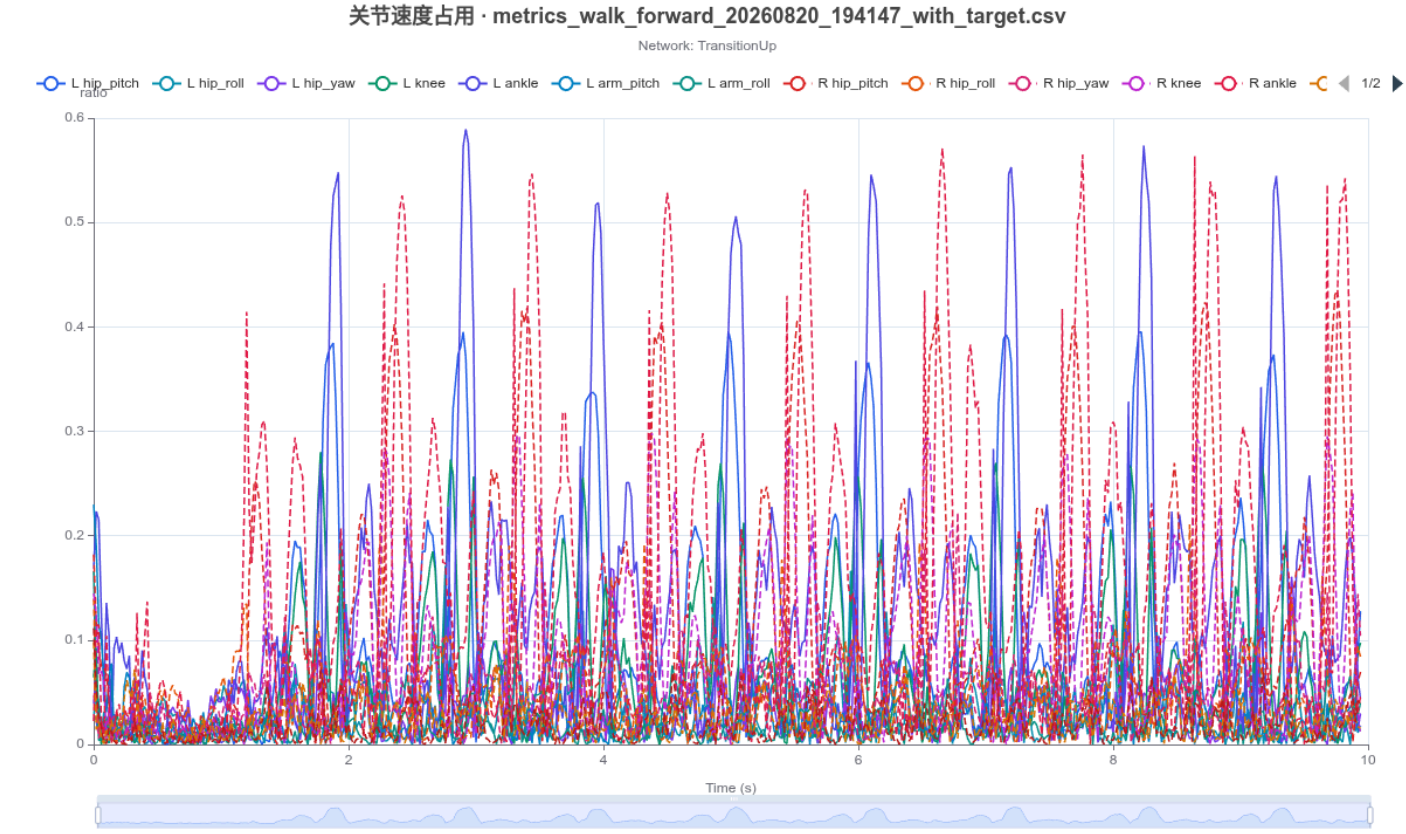


Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
joint_torque_ratio/left_hip_pitch_joint	30	10	498	0.000168595	0.288988	0.0781429	0.105998	0.288988	0.253498
joint_torque_ratio/left_hip_roll_joint	10	15	498	0.0000107387	0.0838674	0.0188650	0.0249629	0.0838674	0.0503996
joint_torque_ratio/left_hip_yaw_joint	10	15	498	0.0000680463	0.0672062	0.0140780	0.0197098	0.0672062	0.0478409
joint_torque_ratio/left_knee_joint	30	10	498	0.0000960505	0.290768	0.0714662	0.0920901	0.290768	0.180397
joint_torque_ratio/left_ankle_joint	10	15	498	0.000156438	0.137532	0.0207493	0.0292968	0.137532	0.0492663
joint_torque_ratio/left_arm_pitch_joint	10	15	498	0.0000137800	0.0195952	0.00231024	0.00275627	0.0195952	0.00446041
joint_torque_ratio/left_arm_roll_joint	10	15	498	0.00000270325	0.0157044	0.00201501	0.00258019	0.0157044	0.00466972
joint_torque_ratio/right_hip_pitch_joint	30	10	498	0.000735997	0.312709	0.0880126	0.117194	0.312709	0.271907
joint_torque_ratio/right_hip_roll_joint	10	15	498	0.0000367175	0.113498	0.0248188	0.0346569	0.113498	0.0769977
joint_torque_ratio/right_hip_yaw_joint	10	15	498	0.0000327835	0.0910654	0.0187854	0.0271423	0.0910654	0.0623075
joint_torque_ratio/right_knee_joint	30	10	498	0.000436727	0.333822	0.0840912	0.104035	0.333822	0.208324
joint_torque_ratio/right_ankle_joint	10	15	498	0.0000244965	0.149578	0.0217372	0.0313262	0.149578	0.0675842
joint_torque_ratio/right_arm_pitch_joint	10	15	498	0.0000138352	0.0110724	0.00209701	0.00245695	0.0110724	0.00420829
joint_torque_ratio/right_arm_roll_joint	10	15	498	0.00000535702	0.0111612	0.00191741	0.00233120	0.0111612	0.00446008

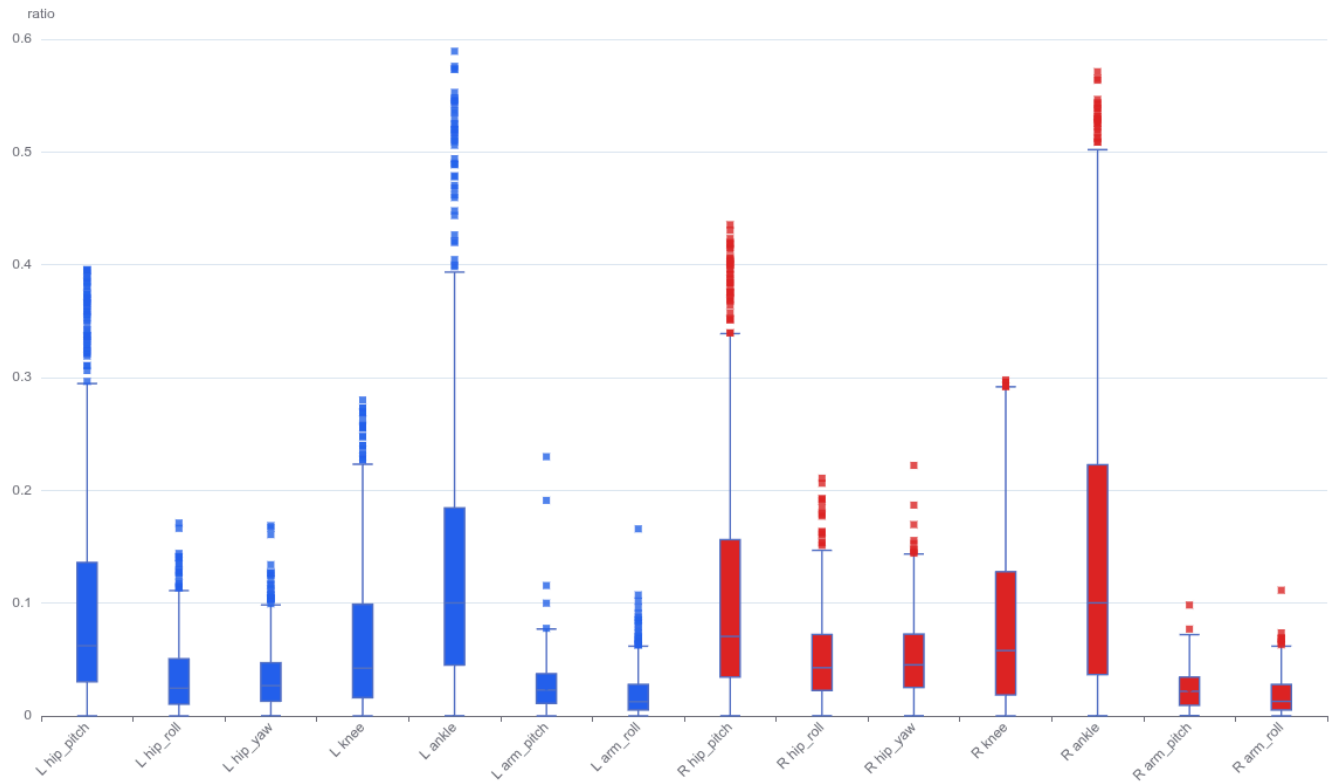

ratio
0.35
0.3
0.25
0.2
0.15
0.1
0.05
0
L hip_pitch
L hip_roll
L hip_yaw
L knee
L ankle
L arm_pitch
L arm_roll
R hip_pitch
R hip_roll
R hip_yaw
R knee
R ankle
R arm_pitch
R arm_roll

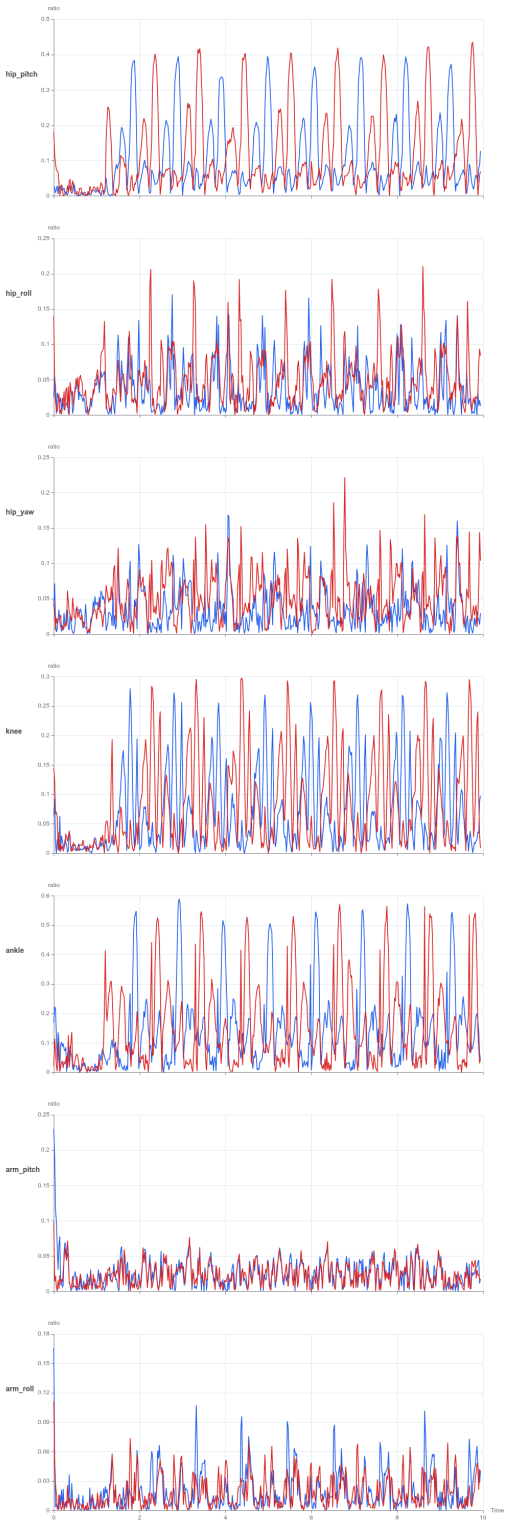

hip_pitch
hip_roll
hip_yaw
knee
ankle
arm_pitch
arm_roll
Time

关节速度占用

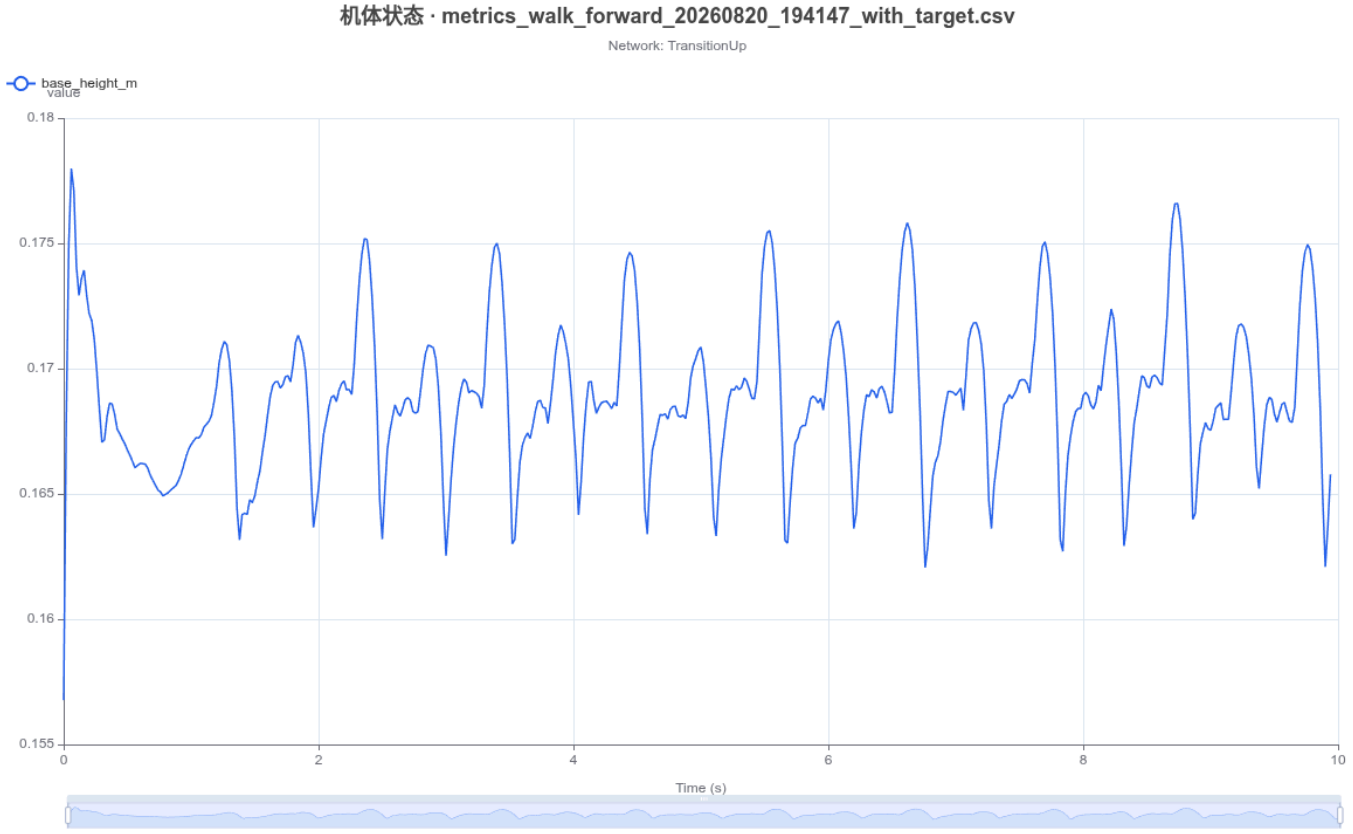


Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
joint_velocity_ratio/left_hip_pitch_joint	30	10	498	0.0000332101	0.395812	0.101139	0.144257	0.395812	0.353319
joint_velocity_ratio/left_hip_roll_joint	10	15	498	0.00000840432	0.171068	0.0355514	0.0483738	0.171068	0.105700
joint_velocity_ratio/left_hip_yaw_joint	10	15	498	0.000197876	0.168712	0.0344424	0.0449938	0.168712	0.0935871
joint_velocity_ratio/left_knee_joint	30	10	498	0.0000387455	0.280117	0.0694098	0.0972503	0.280117	0.212432
joint_velocity_ratio/left_ankle_joint	10	15	498	0.000185240	0.589308	0.140586	0.193939	0.589308	0.488810
joint_velocity_ratio/left_arm_pitch_joint	10	15	498	0.00000260467	0.229832	0.0259816	0.0334871	0.229832	0.0570766
joint_velocity_ratio/left_arm_roll_joint	10	15	498	0.0000348408	0.165865	0.0197538	0.0284807	0.165865	0.0599374
joint_velocity_ratio/right_hip_pitch_joint	30	10	498	0.000138557	0.435542	0.113119	0.158138	0.435542	0.380130
joint_velocity_ratio/right_hip_roll_joint	10	15	498	0.000195498	0.210765	0.0508765	0.0639888	0.210765	0.127849
joint_velocity_ratio/right_hip_yaw_joint	10	15	498	0.000121762	0.222108	0.0514188	0.0623357	0.222108	0.118193
joint_velocity_ratio/right_knee_joint	30	10	498	0.0000307404	0.297957	0.0851408	0.115387	0.297957	0.240977
joint_velocity_ratio/right_ankle_joint	10	15	498	0.000122616	0.571162	0.150033	0.208415	0.571162	0.495096
joint_velocity_ratio/right_arm_pitch_joint	10	15	498	0.000359022	0.0983233	0.0236437	0.0286293	0.0983233	0.0531345
joint_velocity_ratio/right_arm_roll_joint	10	15	498	0.0000113394	0.111452	0.0180920	0.0243465	0.111452	0.0490675

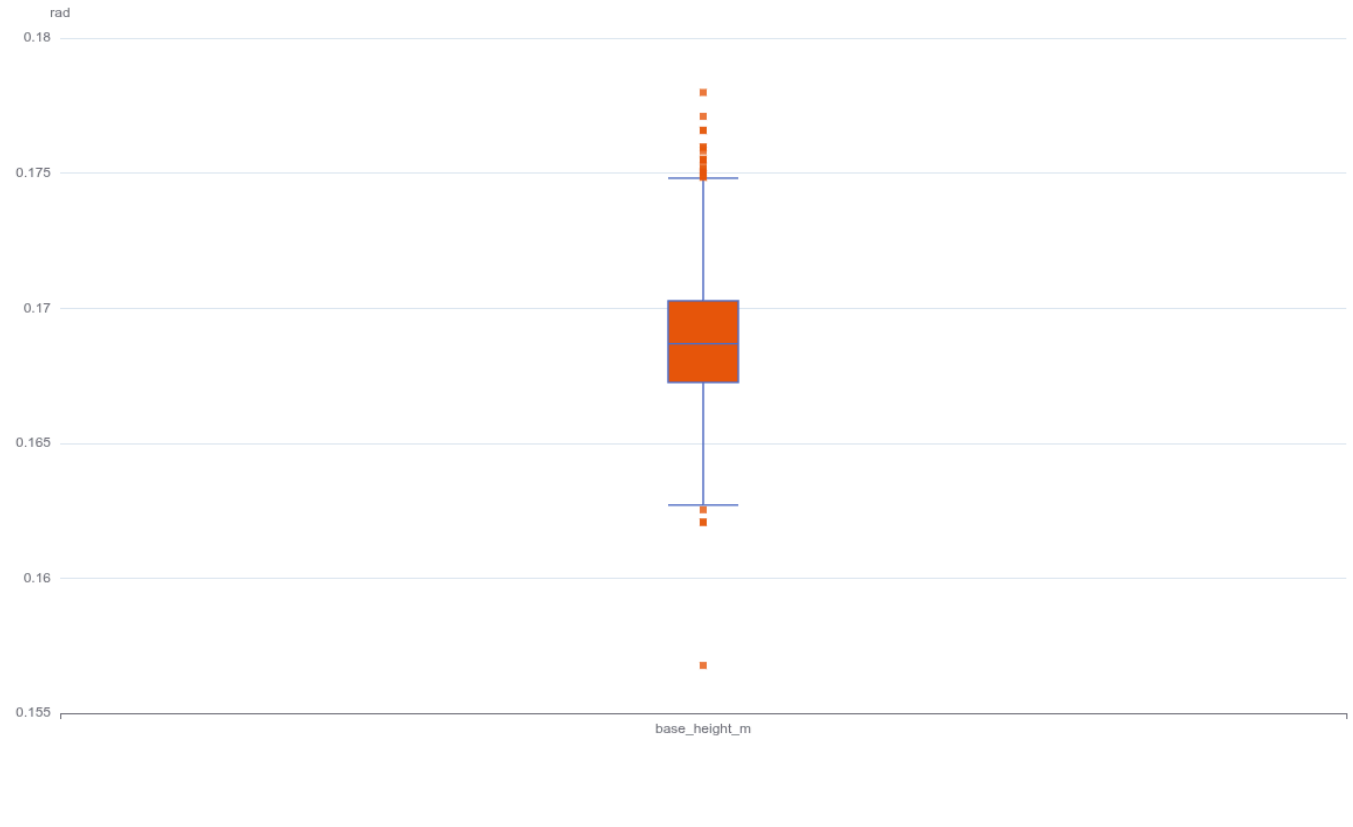

ratio
0.6
0.5
0.4
0.3
0.2
0.1
0
L hip_pitch
L hip_roll
L hip_yaw
L knee
L ankle
L arm_pitch
L arm_roll
R hip_pitch
R hip_roll
R hip_yaw
R knee
R ankle
R arm_pitch
R arm_roll


hip_pitch
hip_roll
hip_yaw
knee
ankle
arm_pitch
arm_roll
ratio
Time

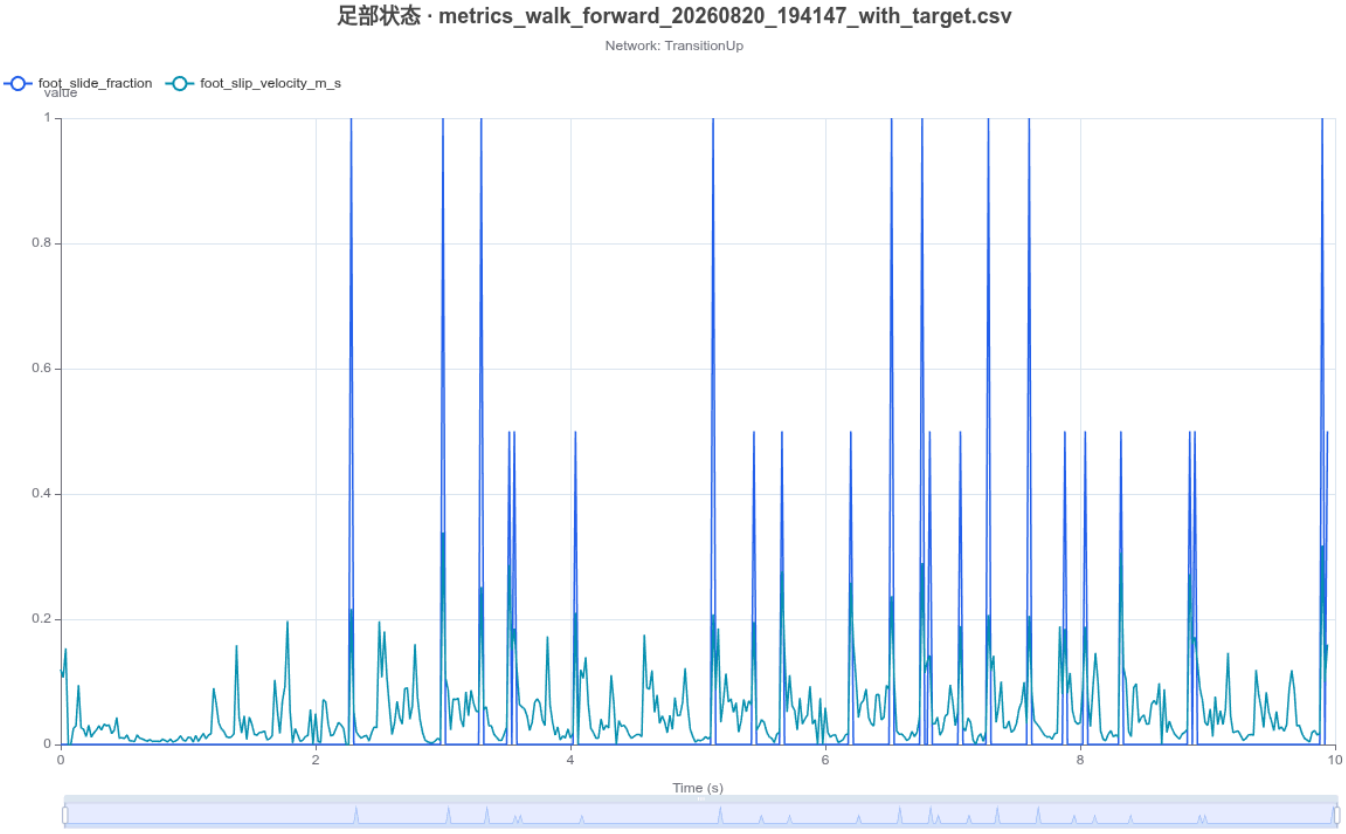
机体状态



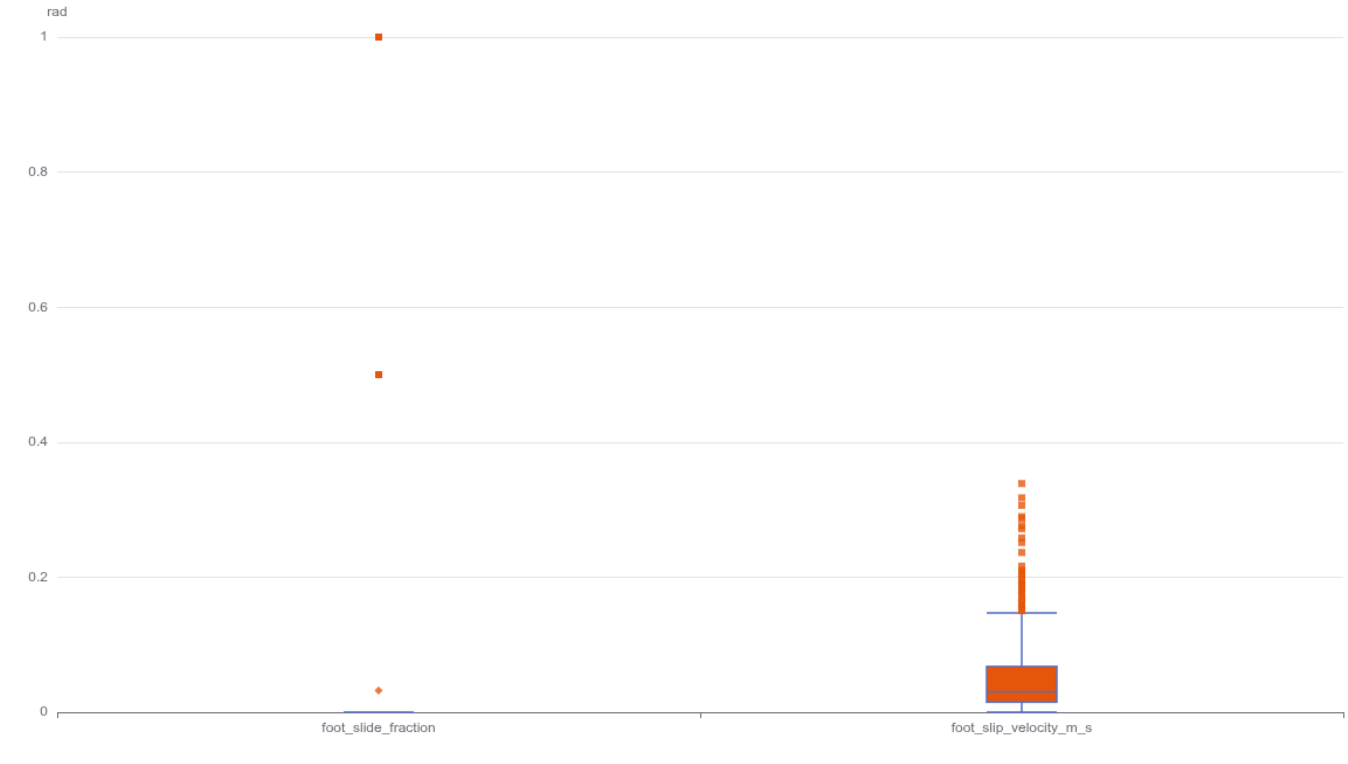
Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
base_height_m	–	–	498	0.156768	0.177987	0.168867	0.168893	0.177987	0.174649



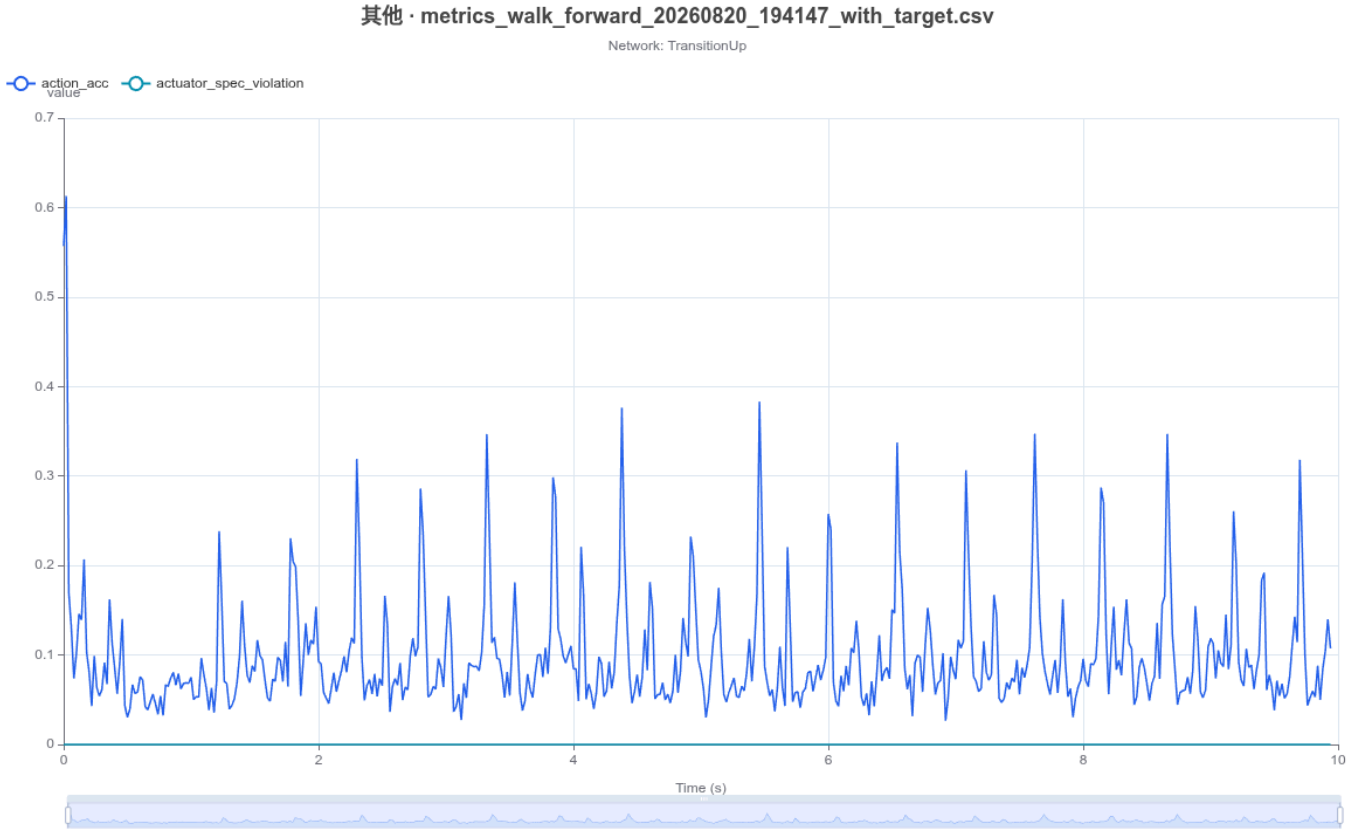
足部状态



Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
foot_slide_fraction	–	–	498	0.00000	1.00000	0.0321285	0.158431	1.00000	0.00000
foot_slip_velocity_m_s	–	–	498	0.00000	0.338640	0.0508822	0.0757865	0.338640	0.175090



其他



Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
action_acc	–	–	498	0.0265813	0.612897	0.0993694	0.119358	0.612897	0.230423
actuator_spec_violation	–	–	498	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000

